YU FU

HUMAN-AI INTERACTION | ADAPTIVE AI AGENTS | HUMAN-CENTERED AI

fuyu1391881922@gmail.com | Sichuan University-Pittsburgh Institute
Google Scholar (60+ citations) | Personal Website

Education

Sichuan University-Pittsburgh Institute *Sep 2023 - Jun 2027 (expected)*

B.Eng. in Computer Science and Technology | GPA: **3.84/4.00**
Honors: National Scholarship (2024); First-Class University Scholarship (2025); Dean's List

Tsinghua University *Fall 2027*

Incoming M.S. in Data Science and Information Technology

Selected Research Projects

ECHO: Explainable Co-editing for Presentation Slide Refinement *First Author; ACM ICMI 2026*

- Designed a multimodal slide co-editing system that grounds natural-language requests in selected objects and converts intent into inspectable, schema-constrained edit plans. Implemented Plan-Confirm-Execute, stable object grounding, adaptive LLM/VLM routing, style memory, and rollback.
- Built CoEdit-Eval and evaluated with 14 participants; task-completion time decreased by **25.8%** and NASA-TLX workload by **20.8%** versus the baseline.

AICo & DefenseMitigator: AI Counseling Systems *CHI 2027 / UIST 2026 Poster*

- For DefenseMitigator (third author), contributed to adaptive, evidence-grounded guidance using a psychological-defense knowledge graph and BDI model; synthesized formative studies with 6 counselors and 4 clients into autonomy-preserving interaction requirements.
- AICo extends this work to expert-supervised, multi-client collaboration. A three-week deployment involved 6 licensed psychological experts, each overseeing 5 parallel conversations; behavioral records and interviews examined supervision and intervention timing. Submitted to CHI 2027.

MathTrace: Process-Aware Analysis of Mathematical Reasoning *AI for Education / HCI*

- Co-designed pilots and analyzed timestamped smart-pen traces (pauses, revisions, backtracking, scratch-to-solution transitions); developed a replay/diagnostic dashboard linking handwriting events to step-level evidence and feedback.

AgentMV: State-Guided Multi-Agent Music Video Generation *Fifth Author; AI4Art @ ECCV 2026*

- Contributed to a long-horizon framework coordinating specialist agents for music understanding, script planning, asset creation, segment generation, and final assembly through a Structured Persistent State.

CassMem: Knowledge-Graph Memory for Long-Term AI Agents *Core Researcher; submitted to WWW 2027*

- Developed Cassandra-backed graph memory with structured/raw representations, hybrid BM25+dense retrieval, and version-aware readiness checks. On 1,146 held-out LoCoMo queries, Z-score fusion achieved **MRR@10 = 0.550** versus 0.523 for compact RRF.

Research & Industry Experience

Tsinghua University, Pervasive Interaction Lab *Research Intern; Mar-Aug 2026*

- Developed a digital-employee agent for graduate-admissions consulting, integrating automated WeChat-group replies with Feishu conversation records and human-expert workflows; supported latency monitoring and conversation-quality review.

Ernst & Young (EY) *AI & Data Intern; Jan-Mar 2026*

- Contributed to invoice recognition and Text-to-Slides across product research, solution evaluation, implementation, and integration.

Selected Publications

Fu, Y., Kang, Y., Zhou, Y., Zhao, Y. *ECHO: Explainable Co-editing with Human-in-the-loop Operations for Presentation Slide Refinement.* ACM ICMI 2026, pp. 99-107. doi:10.1145/3776574.3831120.

Wu, Z., Zhang, R., Fu, Y., Kang, Y., Zhao, Y. *Automated Benchmark Construction for LLM Evaluation: A Source-Generator-Validator-Lifecycle Framework.* PRICAI 2026, accepted.

Nie, Z., Li, H., Fu, Y., Zhang, S., Lu, J., Yu, Z., Cai, J. *From Defense to Disclosure: Designing Interactive Systems with Multimodal Adaptive Guidance for LLM-Based Mental Health Counseling.* ACM UIST 2026 Poster, accepted.

Wang, H., Ouyang, L., Xia, C., Kang, Y., Fu, Y., Ouyang, Y. *AgentMV: A State-Guided Multi-Agent Framework for Budget-Aware Music Video Generation.* AI4Art @ ECCV 2026, accepted.

Fu, Y., Cai, L., Wu, R., Kang, Y., Zhao, Y. *Nutrient-Aware Personalized Meal Recommendation Using Structured Food Knowledge and Constraint Verification.* Foods, 15(10), 1647, 2026. doi:10.3390/foods15101647.

Technical Skills

Programming: Python, C/C++, Java, SQL **AI:** LLM/VLM APIs, RAG, knowledge graphs, agent workflows, hybrid retrieval, evaluation

HCI & Evaluation: Interactive prototyping, formative studies, semi-structured interviews, user-study design, qualitative/quantitative analysis

Tools: PyTorch, FastAPI, Cassandra, Neo4j, Git **English:** IELTS 7.0